Swiss CPI inflation forecast, 2026Q4–2029Q3

Martin Sinishta, Yanik Bittel, Evan Weihs · Meeting of 4 November 2026 · Details in the technical report

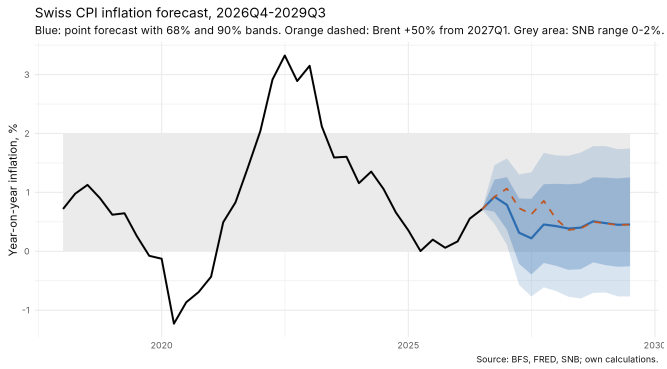
Where we are. Headline inflation is forecast at **0.9% in 2026Q4**, settling at **about 0.45%** from mid-2027. A permanent Brent +50% shock adds **+0.4 pp for one year**. Out of sample (55 origins since 2012) the forecasts are **unbiased** and **beat a headline ARIMA and a random walk at 2–3 years**, but are **not efficient beyond one year**.

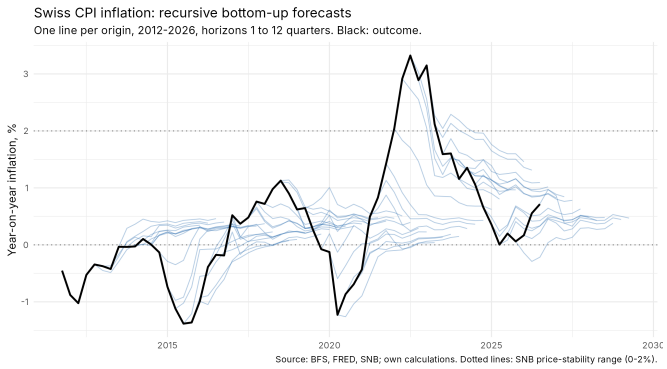
1. Models: estimated on 2000Q1–2026Q3, $y = 100\ln(\text{index})$

Block	Weight	Model	Key estimates	
Domestic excl. rents	59%	ARIMA(0,1,0)(1,0,0)$_4$	drift 0.6%/yr; seasonal AR 0.73; Ljung–Box $p = 0.11$	**OK**
Imported excl. oil	20%	ARIMA(2,1,1)(1,0,0)$_4$	drift −0.6%/yr (n.s.); AR root −0.98 cancels MA root −0.74	**Issue**
Rents	19%	ARIMA(1,1,1)(1,0,0)$_4$	drift 1.4%/yr; all coefficients significant; LB $p = 0.74$	**OK**
Oil products	2%	ECM	long run: Brent 0.43, CHF/USD 0.61, trend 1.7%/yr; 39% of gap closed per quarter	**OK**

Checks: optimal-forecast recursions coded from our derivations = R's `predict()` (to 10^{-13}); blocks re-aggregate to headline within 0.14 pp; Engle–Granger cointegration yes, Johansen no (same long-run vector).

2. Forecast and out-of-sample record





Horizon h (quarters)	1	4	8	12
RMSE, pp	0.36	1.06	1.29	1.20
Unbiased (p)	0.71	0.90	0.84	0.50
Mincer–Zarnowitz (p) / strong efficiency (p)	0.65 / 0.01	0.36 / 0.04	0.01 / 0.01	0.00 / 0.00
MSFE ratio vs headline ARIMA (DM p)	0.99 (0.97)	1.04 (0.80)	**0.80** (0.04)	**0.75** (0.04)
MSFE ratio vs random walk (DM p)	0.78 (0.29)	0.84 (0.39)	**0.56** (0.02)	0.46 (0.07)

Reading: no bias; disaggregation pays off at 2–3 years; beyond one year the model extrapolates the current situation and misses turning points (2022 surge, 2024–25 fall), so efficiency is rejected.

3. Code

`main.R`	All parameters in one place; runs the pipeline; one run (30 s) reproduces every table and figure
`01–02`	Data from APIs (BFS, FRED, SNB), never edited by hand; block indices built per annual chain link; quarterly averages
`04_model_estimation`	`stats::arima` (ML), `lm`, `urca`; no auto-selection; diagnostics, cointegration, checks
`05_forecasting`	Hand-coded recursions `arima_paths()`, `ecm_paths()`; 5,000 joint simulated paths; scenario
`06_evaluation`	Recursive loop re-estimating at 55 origins; own MZ, efficiency and DM tests with Newey–West errors

4. Decisions we need from you

Each point: what we propose, why, and the alternative.

1. **Seasonality.** The non-oil models have a seasonal AR term (so SARIMA). *We propose:* test for seasonality, adjust with X-13 where present, then plain ARIMA (TA's advice); the y-y target is unaffected. *Alternative:* keep the seasonal AR term and justify it.

   Your answer: _____

2. **Cointegration.** Engle–Granger: $-5.30 < -4.10$ (cointegrated). Johansen trace: $37.98 < 42.44$ (not). *We propose:* keep the ECM, report both; the two give the same long-run vector. *Alternative:* ARDL bounds test.

   Your answer: _____

3. **ECM trend.** Oil products drift up 1.7%/yr for given Brent and CHF/USD; with random-walk drivers the oil forecast is the trend alone after a year. *We propose:* keep it (taxes, CO_2 levy). *Alternative:* CO_2-levy step dummies instead of a trend.

   Your answer: _____

4. **Brent and CHF/USD.** *We propose:* random walk, plus a Brent shock scenario for the SNB. *Alternative:* the futures curve (market-based, but no free API).

   Your answer: _____

5. **Imported block.** The negative drift (-0.14/quarter, n.s., from the 2009–16 strong franc) lowers headline by 0.13 pp over the whole horizon; AR and MA roots nearly cancel. *We propose:* drop the drift and estimate ARIMA(1,1,0) or (0,1,1). *Alternative:* 2009–2016 dummy, or a later sample start.

   Your answer: _____

6. **Density bands.** The 90% half-width at $h = 8$ is 1.24 pp, while the out-of-sample RMSE implies about 2.1 pp. *We propose:* evaluate coverage, then calibrate the bands on out-of-sample errors. *Alternative:* add parameter uncertainty by bootstrap.

   Your answer: _____

7. **Weights before 2016.** Recovered from the CPI identity by least squares (error ≤ 0.04 pp against published 2016–26 weights). *We propose:* use them. *Alternative:* published series only, from 2016.

   Your answer: _____

5. Next steps

- Script 03: unit-root tests (deterministic vs stochastic trend for domestic), seasonality plots, X-13.
- Re-estimate the non-oil blocks as plain ARIMA; simplify the imported model; additive outliers for COVID and 2022.
- Density forecast evaluation (coverage, PIT); rerun the evaluation with the revised models.
- Minto pyramid of the statement; technical and non-technical reports.

Notes: